VOLUME 8 — MODEL PAPER 3 (Difficulty: Moderate → Difficult)

NID DAT (B.Des) Prelims: Practice Model Paper 3
Handbook-scheme format. This practice paper was written for this package; it is not an official NID paper.
Focus mix: art history, harder spatial reasoning, expressive typography + logo, bird's-eye storytelling, packaging with dieline, service/system design.

INSTRUCTIONS

1. Time: **3 hours**. Total: **100 marks**. No negative marking.
2. Part I: Q1–14 (1 mark each), Q15–22 (2 marks each) = 30.
3. Part II: Q23 (15), Q24 (20), Q25 (15), Q26 (20) = 70.
4. Suggested timing: Part I = 25 min · Q23 = 25 · Q24 = 40 · Q25 = 28 · Q26 = 42 · buffer = 20.

PART I — OBJECTIVE (30 marks)

Q1–Q14 (1 mark each)

Q1. Which artist led Fauvism, known for wild, non-natural colour?
(a) Claude Monet (b) Henri Matisse (c) Salvador Dalí (d) Edvard Munch

Q2. Patola saris of Patan are unique because they are:
(a) block printed (b) double ikat (c) embroidered with mirrors (d) woven with zari only

Q3. What is the angle between the hour and minute hands of a clock at 3:40?
(a) 120° (b) 130° (c) 140° (d) 150°

Q4. Which is NOT one of Le Corbusier's "Five Points of Architecture"?
(a) Pilotis (b) Roof garden (c) Horizontal ribbon windows (d) Flying buttress

Q5. Which word reads exactly the same when the page is turned upside down (rotated 180°)?
(a) NOON (b) MOON (c) SOON (d) BOON

Q6. How many edges does a regular octahedron (8 triangular faces, 6 vertices) have?
(a) 8 (b) 10 (c) 12 (d) 14

Q7. Srikalahasti-style Kalamkari is made mainly by:
(a) block printing (b) freehand drawing with a bamboo pen (c) tie-dye (d) machine printing

Q8. Odd one out: Futura, Gill Sans, Bodoni, Helvetica
(a) Futura (b) Gill Sans (c) Bodoni (d) Helvetica

Q9. A palette of blue, red-orange and yellow-orange is a:
(a) triadic scheme (b) split-complementary scheme (c) analogous scheme (d) monochromatic scheme

Q10. The Konark Sun Temple was built by:
(a) Rajaraja Chola I (b) Narasimhadeva I (c) Krishnadeva Raya (d) Raja Bhoja

Q11. 1, 3, 6, 10, 15, ?
(a) 20 (b) 21 (c) 22 (d) 25

Q12. The striped IBM logo is still read as solid letters because of the Gestalt principle of:
(a) Proximity (b) Closure (c) Common fate (d) Figure–ground reversal

Q13. Bauhaus : Germany :: De Stijl : ?
(a) France (b) Netherlands (c) Italy (d) Russia

Q14. Aluminium is widely used for beverage cans mainly because it is:
(a) heavy and magnetic (b) lightweight, corrosion-resistant and highly recyclable (c) transparent (d) biodegradable

Q15–Q22 (2 marks each)

Q15. A triangle has 4 straight lines drawn from its top vertex to its base (the two sides plus two internal lines). Two horizontal lines are then drawn across the triangle, parallel to the base and crossing all 4 lines. How many triangles are there in total?
(a) 12 (b) 15 (c) 18 (d) 24

Q16. An 8 × 8 square sheet is folded along the diagonal joining the top-left corner (0, 8) to the bottom-right corner (8, 0), so that the bottom-left triangle lies on the top-right triangle. A hole is punched at (7, 5). After unfolding, where is the second hole?
(a) (5, 7) (b) (1, 3) (c) (3, 1) (d) (7, 3)

Q17. A 2 × 3 × 4 cuboid is painted on all faces and cut into 24 unit cubes. How many unit cubes have exactly ONE face painted?
(a) 0 (b) 4 (c) 8 (d) 12

Q18. In a design competition, A scored more than B but less than C. D scored less than B. E scored more than C. Who came third?
(a) C (b) A (c) B (d) E

Q19. A label wraps exactly once around a cylindrical tin of radius 7 cm and height 10 cm, with 1 cm extra for overlap. What size should the label be? (π = 22/7)
(a) 44 cm × 10 cm (b) 45 cm × 10 cm (c) 22 cm × 10 cm (d) 154 cm × 10 cm

Q20. In a large hospital, first-time visitors keep getting lost. Which solution is MOST effective?
(a) More security guards
(b) Colour-coded floor lines leading to departments, with consistent pictogram signs at every decision point
(c) A long text notice at the entrance
(d) Brighter wall paint everywhere

Q21. Which statement is correct?
(a) Kerning adjusts the space across a whole line; tracking adjusts one letter pair
(b) Kerning adjusts the space between a specific letter pair; tracking adjusts it uniformly across a word or line
(c) Both terms mean line spacing
(d) Leading adjusts the letter width

Q22. A 15-tooth gear drives a 45-tooth gear. How many turns must the small gear make for the large gear to turn 4 times?
(a) 4 (b) 9 (c) 12 (d) 15

PART II — SUBJECTIVE (70 marks)

Q23. Expressive Typography + Logo (15 marks)
(a) Write each of the words **MELT, FLY, CRACK** and **ECHO** so that its appearance expresses its meaning (8 marks).
(b) Design a logo for a monsoon festival called **"BAARISH"**. Show it in colour and in single-colour (black) versions (7 marks).

Q24. Bird's-Eye Storytelling (20 marks)
A pigeon sits on the roof girder of a busy railway station and looks down. Draw what it sees in **bird's-eye perspective**. Within the crowd, include **one small story** the pigeon notices (for example, someone helping, a reunion, a chase). Make the story findable but not obvious.

Q25. Packaging with Dieline (15 marks)
Design **eco-friendly packaging for six kulhads** (clay cups) to be sent by courier. Show: (a) the flat net (dieline) with fold and cut lines, (b) the assembled package, (c) how each kulhad is protected from breaking, and (d) a simple graphic identity on the box.

Q26. Service / System Design (20 marks)
At a very large festival gathering (like the Kumbh Mela), many people, especially children and the elderly, get separated from their families.
(a) Identify **three user groups** and one problem for each (4 marks).
(b) Design a **lost-and-found system**: include an identification idea (e.g. a wristband or token), a booth design and signage (10 marks).
(c) Show in a **3-frame storyboard** how a lost child is reunited using your system (6 marks).

ANSWER KEY — PART I

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	7	b	13	b	19	b
2	b	8	c	14	b	20	b
3	b	9	b	15	c	21	b
4	d	10	b	16	c	22	c
5	a	11	b	17	b		
6	c	12	b	18	b		

DETAILED SOLUTIONS — PART I

Q1 → (b) Matisse. Monet = Impressionism; Dalí = Surrealism; Munch = Expressionism. *Diff:* E.

Q2 → (b) double ikat. Both the warp and weft threads are resist-dyed before weaving, so the pattern appears on both sides. *Diff:* M.

Q3 → (b) 130°. $\theta = |30H - 5.5M| = |90 - 220| = 130°$. *Diff:* M.

Q4 → (d) Flying buttress. That is a Gothic element. The five points are pilotis, free plan, free façade, horizontal windows and roof garden. *Diff:* M.

Q5 → (a) NOON. A 180° rotation reverses the letter order and turns each letter upside down. N and O both look the same upside down, and NOON is the same in reverse, so it reads NOON again. MOON becomes NOOW, because M turns into W. SOON becomes NOOS, and the B in BOON doesn't survive. *Diff:* H.

Q6 → (c) 12. Euler's formula V − E + F = 2 gives 6 − E + 8 = 2, so E = 12. *Diff:* M.

Q7 → (b) freehand pen. Machilipatnam Kalamkari is the block-printed style. *Diff:* M.

Q8 → (c) Bodoni. It is a high-contrast serif (Didone) typeface; the other three are sans serif. *Diff:* M.

Q9 → (b) split-complementary. The complement of blue is orange; its neighbours red-orange and yellow-orange are used instead. *Diff:* M.

Q10 → (b) Narasimhadeva I of the Eastern Ganga dynasty, 13th century. Rajaraja Chola I built Thanjavur's Brihadeeswara. *Diff:* M.

Q11 → (b) 21. These are triangular numbers, with differences +2, +3, +4, +5, +6. *Diff:* E.

Q12 → (b) Closure. The mind fills the gaps between the stripes. *Diff:* M.

Q13 → (b) Netherlands. *Diff:* E.

Q14 → (b). Aluminium is light, does not rust and can be recycled repeatedly with large energy savings. *Diff:* E.

Q15 → (c) 18. Each triangle uses 2 of the 4 apex lines plus one horizontal "base". There are C(4,2) = 6 pairs of lines and 3 bases (the 2 crossing lines + the actual base), so 6 × 3 = 18. *Diff:* H.

Q16 → (c) (3, 1). The fold line is x + y = 8. Reflecting (x, y) across it gives (8 − y, 8 − x) = (8 − 5, 8 − 7) = (3, 1). *Check:* the midpoint of (7,5) and (3,1) is (5,3), which lies on x + y = 8 ✓. The trap is (5, 7), which comes from swapping coordinates as if the fold were y = x. *Diff:* H.

Q17 → (b) 4. One-face cubes = 2[(a−2)(b−2) + (b−2)(c−2) + (a−2)(c−2)] with a = 2, b = 3, c = 4. That is 2[0 × 1 + 1 × 2 + 0 × 2] = 2 × 2 = 4. *Check:* corners 8 + two-face 4[(0) + (1) + (2)] = 12 + one-face 4 + inside 0 = 24 ✓. *Diff:* H.

Q18 → (b) A. The order is E > C > A > B > D. *Diff:* M.

Q19 → (b) 45 cm × 10 cm. Circumference = 2 × 22/7 × 7 = 44 cm; adding 1 cm of overlap gives 45 cm. Height 10 cm. *Diff:* M.

Q20 → (b). A continuous guiding cue (floor lines) plus signs at the decision points matches how people navigate. (c) overloads memory; (a) and (d) don't solve wayfinding. *Diff:* M.

Q21 → (b). *Diff:* E.

Q22 → (c) 12. The large gear's 4 turns pass 4 × 45 = 180 teeth; 180 ÷ 15 = 12 turns of the small gear. *Diff:* M.

DETAILED SOLUTIONS — PART II

Q23. Expressive typography + BAARISH logo (15 marks)

(a) Model treatments (2 marks each)

- **MELT:** bold letters whose lower halves drip and pool into a puddle; the "T" bends.

- **FLY:** light letters rising diagonally upward; the "Y" becomes a bird's wing shape, with small motion lines.
- **CRACK:** solid block letters split by a jagged crack running through all five; small fragments falling.
- **ECHO:** the word repeated 3–4 times, each repetition lighter and smaller, receding (a fading repeat).

(b) Logo "BAARISH" (7 marks)

- *Concept:* the two **A**s become two **raindrops** (inverted teardrops with the crossbar as a ripple). Alternatively, the dot of the **i** becomes a falling drop that splashes into ripple lines under the word.
- *Colour version:* monsoon teal and deep indigo, with a fresh green accent.
- *Single-colour version:* all black on white, to prove it works without colour.
- *Typeface:* a rounded sans for friendliness.

Rubric: (a) 8 = 4 words × 2 (meaning conveyed + legibility). (b) 7 = concept 3 + execution 2 + mono version working 2.

Mistakes: illegible distortions; simply colouring the words instead of transforming their form; a logo that collapses in black and white.

Q24. Pigeon's-eye railway station (20 marks)

Approach

- **Perspective:** 3-point, looking down. The third VP is below the page, the platforms converge, and we see the tops of heads, umbrellas, luggage, the train roof and the overhead wires in the foreground (the pigeon's girder can frame the top edge).
- **Crowd:** 25+ simplified figures shown as head-and-shoulder ovals from above, with long luggage shadows (for example, late-afternoon light).
- **Hidden story options:**
  - a child's balloon escaping, with someone reaching for it
  - a porter helping an elderly woman with a heavy trunk across the footbridge
  - two friends running towards each other for a reunion
  - a pickpocket being chased by a policeman, shown as a line of displaced people
- **Findability:** place the story at a thirds intersection; give the main characters a single accent colour (e.g. a red dupatta) while the rest of the crowd is muted.

Rubric (20)

Criterion	Marks
Correct bird's-eye perspective	6
Crowd handling & variety	4
Hidden story: clarity & cleverness	4
Composition & use of accent	3
Station details (tracks, platform, signs, train roof)	3

Mistakes: drawing an eye-level view; making the story so central that it isn't "hidden"; repeating identical figures.

Q25. Kulhad packaging (15 marks)

Model concept: "Matti-Pack"

- **Structure:** a corrugated-cardboard box (recycled kraft) with an internal **die-cut honeycomb insert**: 6 conical cut-outs arranged 3 × 2 that cradle each kulhad upside down. The kulhads' rims rest on a **molded-pulp (egg-tray type) bed** or on crumpled newspaper/rice straw.
- **Dieline:** the box net shows solid lines = cut, dashed lines = fold, glue flaps marked; a separate insert net with 6 circular cut-outs and slit tabs that lock into the walls.
- **Assembled view:** 3/4 view with the lid open, showing the kulhads nested.
- **Protection:** each kulhad is isolated (no clay-on-clay contact); there is a 1.5 cm air gap from the walls; a "FRAGILE – this side up" pictogram; no plastic.
- **Graphic identity:** a terracotta-orange band with a simple line icon of a kulhad and steam, a handmade-craft story line ("Made by the potters of Khurja"), and a recycle symbol.
- **Reuse:** the insert doubles as a 6-cup serving tray.

Rubric (15)

Criterion	Marks
Correct, readable dieline	4
Protection logic	4
Sustainability (no plastic, reuse)	3
Graphic identity	2
Clarity of assembled drawing	2

Mistakes: bubble wrap (contradicts eco-friendly); a dieline that cannot fold into the shown box; ignoring breakage risk.

Q26. Festival lost-and-found system (20 marks)

(a) User groups & problems (4)

1. **Young children:** cannot state their address or phone number.
2. **Elderly people:** get confused, may have poor hearing or memory, may not carry a phone.
3. **Rural first-time visitors:** language barriers; unfamiliar with announcements and digital systems.
4. *(optional)* **Volunteers/police:** need quick identification without a common language.

(b) System design (10)

- **ID: "Milan Band"** (milan = reunion). At the entry gates, families receive matching coloured wristbands. Each band carries a family number printed in large digits, plus a QR code linked to the guardian's phone number. A colour indicates the sector of the camp where the family is staying. The bands are tear-resistant Tyvek and waterproof.
- **Booths:** tall, visible "Milan Kendra" towers every 500 m, with a beacon light and a large handshake icon. Each has a seating area, water and a public-address system; volunteers scan the band's QR code, which sends an SMS to the guardian; a loudspeaker announcement is made in 3 languages.
- **Signage:** a consistent icon (two hands joining) in bright yellow-and-black, directional arrows at intersections ("Milan Kendra 200 m →"), and floor stencils.
- **Low-tech backup:** if the phone network fails, families have pre-agreed meeting points (numbered pillars).

(c) 3-frame storyboard (6)

1. A child, crying, is separated in the crowd; a volunteer notices the yellow band.
2. At the Milan Kendra, the volunteer scans the band; the mother's phone buzzes with "Your child is at Milan Kendra 12".

3. A reunion hug at the booth; the band numbers match; a calm environment.

Rubric (20): problems 4; ID design 3; booth 3; signage 2; low-tech/backup thinking 2; storyboard clarity 4; overall feasibility 2.

Mistakes: an app-only solution (ignores the elderly, rural users and network failure); unclear storyboard continuity; no signage.

SCORING INTERPRETATION — MODEL PAPER 3

Practice benchmarks only, not NID cutoffs.

Score	Band
75–100	**Excellent**
60–74	**Strong**
45–59	**Good**
30–44	**Needs improvement**
< 30	**Major improvement required**

Diagnostic tip: Q15–17 are the hardest spatial items in the papers so far. If you missed 2 or more, redo Vol 4B §C–E.

NEXT VOLUME: Volume 9 — Model Paper 4 (`09-VOL9-MODEL-PAPER-4.md`)